

MWD/LWD Tool Operating Ranges & Interpretation Reference

Measurement While Drilling / Logging While Drilling

Permian Basin Horizontal Drilling Operations

1. MWD System Overview

1.1 Mud Pulse Telemetry (Standard Configuration)

- Telemetry type: Positive pulse (mud siren)
- Data rate: 3–12 bits/second (depth-dependent)
- Operating mud weight range: 7.0–20.0 ppg
- Minimum flow rate for telemetry: 150 gpm
- Maximum flow rate: 650 gpm
- Maximum standpipe pressure: 5,000 psi
- Operating temperature: -4°F to 302°F (-20°C to 150°C)
- Pressure rating: 20,000 psi

1.2 Electromagnetic (EM) Telemetry (Alternative — used in air/mist drilling)

- Does not require mud flow
 - Range limited by formation resistivity (max ~8,000 ft in conductive formations)
 - Not recommended for Wolfcamp A laterals >6,000 ft
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2. Directional Sensors

2.1 Survey Accuracy Specifications

Parameter	Standard MWD	High-Accuracy MWD
Inclination accuracy	±0.1°	±0.05°
Azimuth accuracy	±1.0°	±0.5°
Toolface accuracy	±1.5°	±1.0°

Parameter	Standard MWD	High-Accuracy MWD
Survey interval	30–90 ft	30 ft

2.2 Survey Acceptance Criteria (Permian Basin)

- Maximum dogleg severity: 8°/100 ft (build section), 3°/100 ft (lateral)
- Inclination tolerance in lateral: 88°–92° (target: 90°)
- Azimuth tolerance: $\pm 5^\circ$ from planned azimuth
- Closure distance from target: ≤ 50 ft at TD

2.3 Magnetic Interference

- Minimum distance from casing for valid magnetic survey: 50 ft
- If within 50 ft of casing: use gyro survey or non-magnetic drill collars
- Non-magnetic collar length required: minimum 30 ft above and below sensor sub

2.4 Survey Frequency

- Build section: every 30 ft
- Tangent/lateral section: every 60–90 ft
- Near target zone (within 200 ft of formation top): every 30 ft

3. Gamma Ray (GR) Sensor

3.1 Specifications

- Measurement type: Total natural gamma ray (API units)
- Detector: NaI scintillation crystal
- Vertical resolution: 18–24 in
- Depth of investigation: 12–18 in
- Operating temperature: up to 302°F (150°C)
- Logging speed: up to 200 ft/hr (LWD mode)

3.2 Permian Basin Formation GR Signatures

Formation	Typical GR Range (API)	Interpretation
Wolfcamp A shale	80–140 API	Organic-rich shale — target zone
Wolfcamp A carbonate	20–45 API	Carbonate interbeds — avoid if possible
Spraberry siltstone	55–85 API	Moderate GR — target zone
Bone Spring carbonate	15–40 API	Low GR carbonate

Formation	Typical GR Range (API)	Interpretation
Bone Spring shale	90–130 API	High GR shale
Dean sandstone	30–60 API	Low-moderate GR
Wolfcamp B	70–110 API	Below Wolfcamp A target

3.3 Geosteering Using GR

- Landing target: GR > 80 API sustained over 30 ft = confirmed Wolfcamp A shale
- Steer up if: GR drops below 50 API (entering carbonate)
- Steer down if: GR exceeds 150 API (entering organic-rich zone below target)
- Formation top pick: first sustained GR increase >20 API above background

4. Resistivity Sensor (LWD)

4.1 Specifications — ARC (Array Resistivity Compensated)

- Measurement depths: 10, 20, 30, 60, 90 in (5 depths of investigation)
- Frequency: 2 MHz and 400 kHz
- Vertical resolution: 12–24 in (frequency dependent)
- Operating temperature: up to 302°F
- Operating pressure: 20,000 psi

4.2 Resistivity Interpretation — Wolfcamp A

Zone	Resistivity Range (ohm-m)	Interpretation
Organic shale (oil-saturated)	8–25 ohm-m	Target — good oil saturation
Organic shale (water-saturated)	1–4 ohm-m	Avoid — water-bearing
Carbonate stringer	50–500 ohm-m	Hard stringer — potential drilling hazard
Tight siltstone	15–40 ohm-m	Moderate reservoir quality
Brine-saturated zone	0.5–2 ohm-m	Avoid — high water cut risk

4.3 Rt/Rxo Ratio (Invasion Analysis)

- $R_t/R_{xo} > 1.5$: oil-based mud invasion, oil-bearing formation
- $R_t/R_{xo} \approx 1.0$: no invasion contrast (tight formation or water-bearing)

- $R_t/R_{xo} < 0.8$: water-bearing formation (mud filtrate displacing formation water)
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5. Downhole Dynamics Sensors

5.1 Shock & Vibration

- Axial shock limit: 50 g (peak)
- Lateral shock limit: 100 g (peak)
- Torsional vibration (stick-slip) indicator: RPM variation $>\pm 20\%$ of surface RPM
- Recommended action if limits exceeded: reduce WOB by 20%, adjust RPM

5.2 Downhole WOB/Torque

- Downhole WOB sensor accuracy: ± 2 klbs
- Downhole torque sensor accuracy: ± 500 ft-lbs
- Bit bounce indicator: axial vibration > 25 g at > 10 Hz
- Whirl indicator: lateral vibration > 50 g with backward rotation signature

5.3 Annular Pressure While Drilling (APWD)

- Measures ECD directly at sensor depth
 - Sensor location: typically 50–80 ft above bit
 - ECD accuracy: ± 0.05 ppg
 - Use for: real-time pore pressure monitoring, lost circulation detection
 - Alert threshold: $ECD > \text{mud weight} + 0.3$ ppg (potential fracture gradient approach)
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6. Tool Failure Modes & Troubleshooting

6.1 Loss of Telemetry Signal

Cause	Diagnosis	Action
Plugged pulser	No signal at surface, pump pressure normal	Increase flow rate 50 gpm; if no recovery, POOH
Pulser erosion	Weak/intermittent signal	Reduce flow rate; plan tool replacement
Battery depletion	Signal loss after > 200 hrs	POOH for battery replacement
Mud motor bypass	Signal present but no rotation	Check motor differential pressure

6.2 Erratic Survey Data

Symptom	Probable Cause	Action
Azimuth jumps $>5^{\circ}$ between surveys	Magnetic interference	Check proximity to casing; use gyro
Inclination drift $>0.5^{\circ}$ /survey	Accelerometer failure	POOH for tool replacement
Toolface not responding	Stuck motor or bent sub	Check WOB; inspect BHA

6.3 GR Sensor Issues

Symptom	Probable Cause	Action
GR reads zero	Detector failure or cable fault	POOH for replacement
GR reads constant high value	Radioactive tracer in mud	Check mud system; flush if needed
GR spikes	Shock damage to crystal	Reduce vibration; check shock data

7. BHA Design Guidelines — Permian Basin Horizontal

7.1 Typical BHA Configuration (Lateral Section)

1. PDC bit (6-1/8 in)
2. Bit sub
3. Mud motor (6-1/4 in, 1.15° bend)
4. MWD/LWD sub (GR + Resistivity + APWD)
5. Non-magnetic drill collar (30 ft)
6. Jar (hydraulic)
7. HWDP (5 in, 3 joints)
8. Drill pipe to surface

7.2 Motor Specifications (6-1/4 in, 7/8 Lobe)

- Differential pressure at bit: 400–600 psi (normal drilling)
- Maximum differential pressure: 800 psi
- Stall pressure: 900–1,000 psi (do not exceed)
- Motor output RPM: 120–180 RPM at 400 gpm
- Bend setting: 1.15° (standard for Wolfcamp A laterals)
- Maximum dogleg from motor: $8^{\circ}/100$ ft

7.3 Operating Limits Summary

Parameter	Minimum	Maximum	Optimal
Flow rate (gpm)	250	600	400–500
WOB (klbs)	5	35	15–25
Surface RPM	40	120	60–80
Differential pressure (psi)	200	800	400–600
ECD (ppg)	MW	MW + 0.5	MW + 0.2–0.4
Standpipe pressure (psi)	—	4,500	3,000–4,000